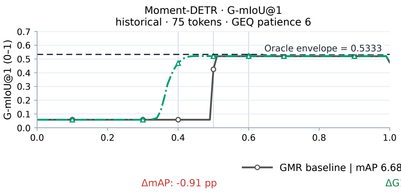
11 backbones · GMR baseline vs GMR + GEQ Full

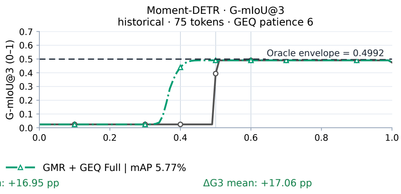
Standard-trained, fixed Val-selected checkpoints → SoccerGMR Standard Test (1036 queries)

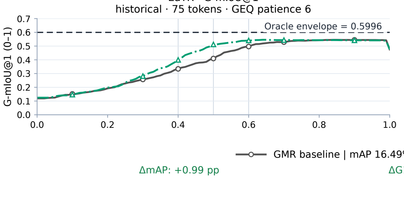
Historical 5: GEQ patience 6, video limit 75 | Expansion 6: GEQ patience 10, video limit 128, seed 2023



GMR baseline | mAP 6.68% GMR + GEQ Full | mAP 5.77%

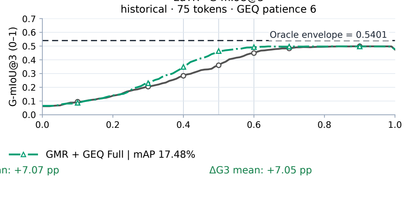
ΔmAP: -0.91 pp ΔG1 mean: +16.95 pp ΔG3 mean: +17.06 pp

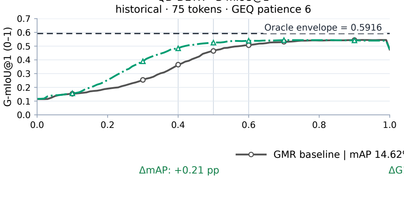




GMR baseline | mAP 16.49% GMR + GEQ Full | mAP 17.48%

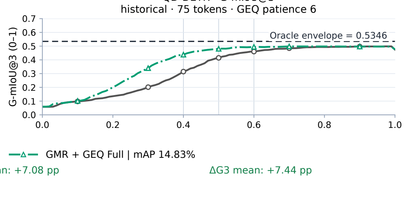
ΔmAP: +0.99 pp ΔG1 mean: +7.07 pp ΔG3 mean: +7.05 pp

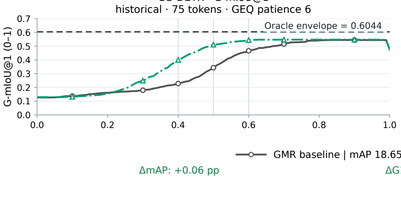




GMR baseline | mAP 14.62% GMR + GEQ Full | mAP 14.83%

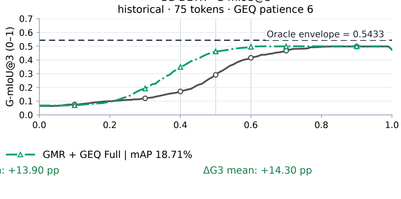
ΔmAP: +0.21 pp ΔG1 mean: +7.08 pp ΔG3 mean: +7.44 pp

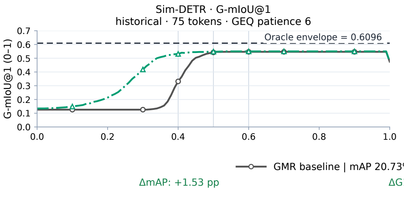




GMR baseline | mAP 18.65% GMR + GEQ Full | mAP 18.71%

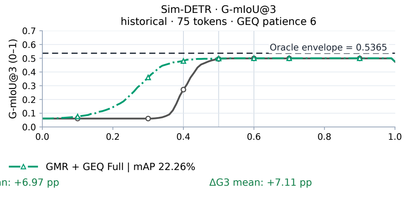
ΔmAP: +0.06 pp ΔG1 mean: +13.90 pp ΔG3 mean: +14.30 pp

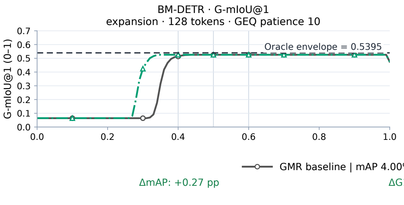




GMR baseline | mAP 20.73% GMR + GEQ Full | mAP 22.26%

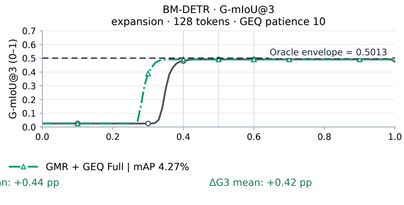
ΔmAP: +1.53 pp ΔG1 mean: +6.97 pp ΔG3 mean: +7.11 pp

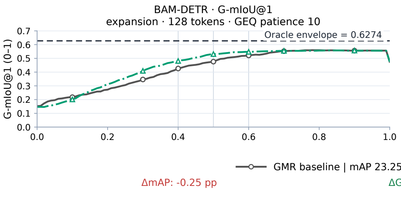




GMR baseline | mAP 4.00% GMR + GEQ Full | mAP 4.27%

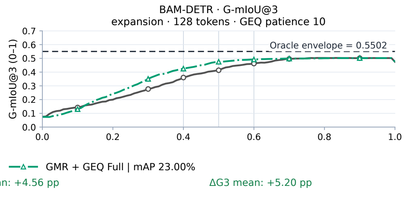
ΔmAP: +0.27 pp ΔG1 mean: +0.44 pp ΔG3 mean: +0.42 pp

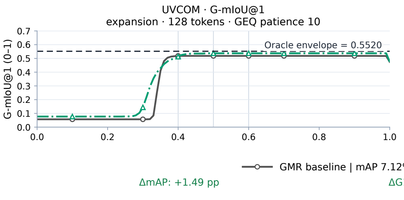




GMR baseline | mAP 23.25% GMR + GEQ Full | mAP 23.00%

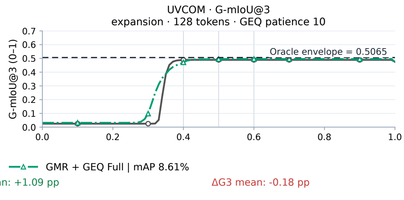
ΔmAP: -0.25 pp ΔG1 mean: +4.56 pp ΔG3 mean: +5.20 pp

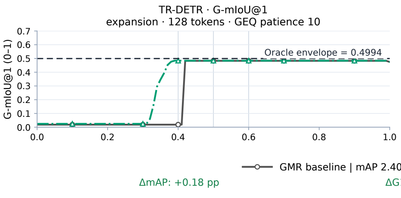




GMR baseline | mAP 7.12% GMR + GEQ Full | mAP 8.61%

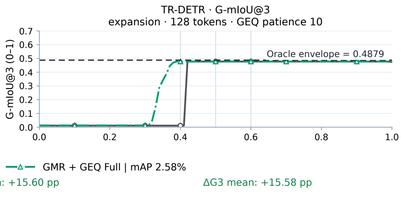
ΔmAP: +1.49 pp ΔG1 mean: +1.09 pp ΔG3 mean: -0.18 pp

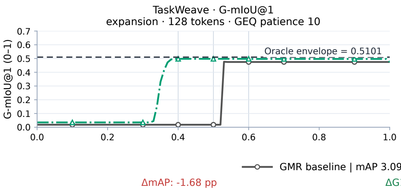




GMR baseline | mAP 2.40% GMR + GEQ Full | mAP 2.58%

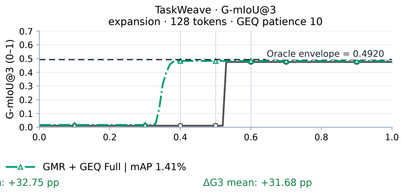
ΔmAP: +0.18 pp ΔG1 mean: +15.60 pp ΔG3 mean: +15.58 pp

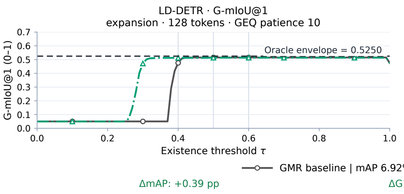




GMR baseline | mAP 3.09% GMR + GEQ Full | mAP 1.41%

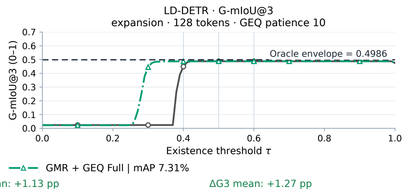
ΔmAP: -1.68 pp ΔG1 mean: +32.75 pp ΔG3 mean: +31.68 pp





GMR baseline | mAP 6.92% GMR + GEQ Full | mAP 7.31%

ΔmAP: +0.39 pp ΔG1 mean: +1.13 pp ΔG3 mean: +1.27 pp



Curves: 101 fixed thresholds, 0–1 units. Δ = GEQ Full − matching GMR; G1/G3 means use τ = .4/.5/.6; pp = percentage points.

Oracle envelope: GT presence with each method's fixed candidates; diagnostic only. Test never selects checkpoints or thresholds.

One seed per experiment; curves show threshold sensitivity, not run uncertainty. Protocol groups are not a controlled cross-backbone ranking.

11 backbones · GMR baseline vs GMR + GEQ Full

Standard-trained, fixed Val-selected checkpoints → SoccerGMR Full-Decodable Test (2970/2986 queries)

Historical 5: GEQ patience 6, video limit 75 | Expansion 6: GEQ patience 10, video limit 128, seed 2023

Moment-DETR · G-mIoU@1

historical · 75 tokens · GEQ patience 6



GMR baseline | mAP 5.47% · GMR + GEQ Full | mAP 5.47%

ΔmAP: +0.00 pp · ΔG1 mean: +15.60 pp · ΔG3 mean: +15.77 pp

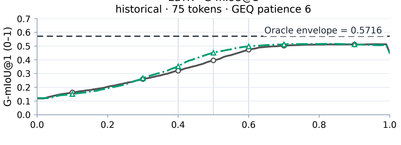
Moment-DETR · G-mIoU@3

historical · 75 tokens · GEQ patience 6



EaTR · G-mIoU@1

historical · 75 tokens · GEQ patience 6

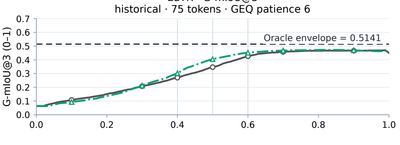


GMR baseline | mAP 15.19% · GMR + GEQ Full | mAP 15.26%

ΔmAP: +0.07 pp · ΔG1 mean: +3.73 pp · ΔG3 mean: +3.84 pp

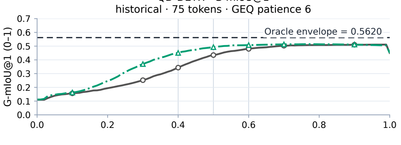
EaTR · G-mIoU@3

historical · 75 tokens · GEQ patience 6



QD-DETR · G-mIoU@1

historical · 75 tokens · GEQ patience 6

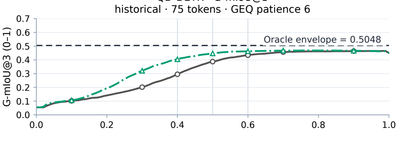


GMR baseline | mAP 12.65% · GMR + GEQ Full | mAP 13.22%

ΔmAP: +0.57 pp · ΔG1 mean: +6.40 pp · ΔG3 mean: +6.59 pp

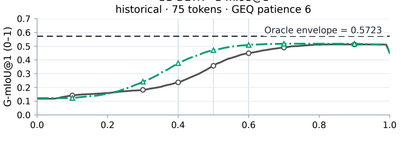
QD-DETR · G-mIoU@3

historical · 75 tokens · GEQ patience 6



CG-DETR · G-mIoU@1

historical · 75 tokens · GEQ patience 6

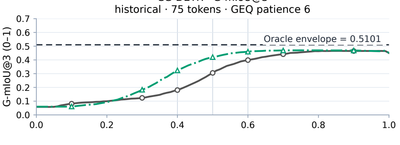


GMR baseline | mAP 14.67% · GMR + GEQ Full | mAP 14.85%

ΔmAP: +0.18 pp · ΔG1 mean: +10.45 pp · ΔG3 mean: +10.65 pp

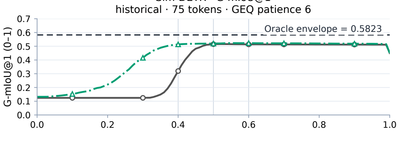
CG-DETR · G-mIoU@3

historical · 75 tokens · GEQ patience 6



Sim-DETR · G-mIoU@1

historical · 75 tokens · GEQ patience 6

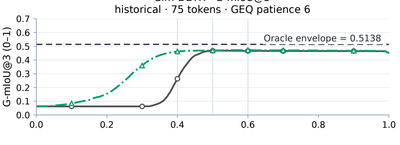


GMR baseline | mAP 19.18% · GMR + GEQ Full | mAP 20.58%

ΔmAP: +1.40 pp · ΔG1 mean: +6.93 pp · ΔG3 mean: +6.90 pp

Sim-DETR · G-mIoU@3

historical · 75 tokens · GEQ patience 6



BM-DETR · G-mIoU@1

expansion · 128 tokens · GEQ patience 10



GMR baseline | mAP 3.65% · GMR + GEQ Full | mAP 4.32%

ΔmAP: +0.67 pp · ΔG1 mean: +0.78 pp · ΔG3 mean: +0.94 pp

BM-DETR · G-mIoU@3

expansion · 128 tokens · GEQ patience 10



BAM-DETR · G-mIoU@1

expansion · 128 tokens · GEQ patience 10



GMR baseline | mAP 20.62% · GMR + GEQ Full | mAP 20.26%

ΔmAP: -0.36 pp · ΔG1 mean: +4.15 pp · ΔG3 mean: +4.56 pp

BAM-DETR · G-mIoU@3

expansion · 128 tokens · GEQ patience 10



UVCOM · G-mIoU@1

expansion · 128 tokens · GEQ patience 10



GMR baseline | mAP 6.40% · GMR + GEQ Full | mAP 8.13%

ΔmAP: +1.73 pp · ΔG1 mean: +0.58 pp · ΔG3 mean: -0.31 pp

UVCOM · G-mIoU@3

expansion · 128 tokens · GEQ patience 10



TR-DETR · G-mIoU@1

expansion · 128 tokens · GEQ patience 10



GMR baseline | mAP 2.36% · GMR + GEQ Full | mAP 2.34%

ΔmAP: -0.02 pp · ΔG1 mean: +14.21 pp · ΔG3 mean: +14.36 pp

TR-DETR · G-mIoU@3

expansion · 128 tokens · GEQ patience 10



TaskWeave · G-mIoU@1

expansion · 128 tokens · GEQ patience 10



GMR baseline | mAP 2.95% · GMR + GEQ Full | mAP 1.33%

ΔmAP: -1.62 pp · ΔG1 mean: +30.16 pp · ΔG3 mean: +29.09 pp

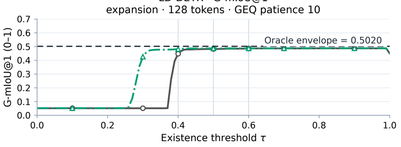
TaskWeave · G-mIoU@3

expansion · 128 tokens · GEQ patience 10



LD-DETR · G-mIoU@1

expansion · 128 tokens · GEQ patience 10

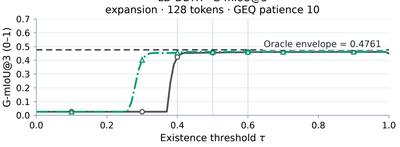


GMR baseline | mAP 6.84% · GMR + GEQ Full | mAP 7.29%

ΔmAP: +0.45 pp · ΔG1 mean: +1.32 pp · ΔG3 mean: +1.33 pp

LD-DETR · G-mIoU@3

expansion · 128 tokens · GEQ patience 10



Curves: 101 fixed thresholds, 0–1 units. Δ = GEQ Full − matching GMR; G1/G3 means use τ = .4/.5/.6; pp = percentage points.

Oracle envelope: GT presence with each method's fixed candidates; diagnostic only. Test never selects checkpoints or thresholds.

Full-Decodable: 2970/2986 queries. Historical 75-token truncation is retained; expansion uses full valid videos (up to 128 tokens).